

Project: Clone Synthesis Tutor (1-Week Challenge)

A 1-Week Challenge to Build a Conversational, AI-Powered Math Tutor

Field	Details
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Problem Statement

Synthesis Tutor has demonstrated the power of combining a conversational AI with interactive, digital manipulatives to teach mathematics effectively. It creates a learning experience that feels like exploration rather than homework, successfully engaging children and building deep conceptual understanding. The core challenge is to replicate this magical experience by building a functional prototype of a single math lesson that captures the essence of the Synthesis model.

Your task is to **build a one-week prototype of a Synthesis-style math tutor**. This involves creating a single, self-contained lesson that uses a conversational interface and an interactive digital manipulative to teach a core elementary math concept.

1-Week Challenge Scope

This is a rapid prototyping sprint. The goal is to build a **single, functional math lesson** that is demonstrable by the end of the week. The focus should be on the quality of the interaction between the tutor and the student, and the effectiveness of the digital manipulative. You are not expected to build a complex, adaptive AI or a full curriculum.

The target concept for this challenge is fractions. You will build a lesson that teaches the concept of fraction equivalence (e.g., that $\frac{1}{2}$ is the same as $\frac{2}{4}$).

Functional Requirements

1. Conversational Tutor Interface

- The application must feature a chat-style interface where a “tutor” provides instruction and asks questions.
- The tutor’s dialogue should be **scripted** for this 1-week challenge. You do not need to implement a complex LLM-based conversational agent.
- The script should guide the student through the lesson, respond to correct and incorrect answers with simple branching logic, and have a warm, encouraging tone.

2. Interactive Digital Manipulative

- The application must include an interactive workspace where the student can manipulate digital representations of fractions.
- This should include a “fraction box” or similar visual tool where students can see and interact with fraction blocks (e.g., a block representing $\frac{1}{2}$, two blocks representing $\frac{1}{4}$, etc.).
- The student must be able to perform actions like combining, splitting, or “smashing” fraction blocks to understand their relationships.

3. Lesson Flow

- The lesson must guide the student through the concept of fraction equivalence.
- It should start with exploration using the manipulatives, followed by more formal questions from the tutor.
- The lesson should conclude with a simple “check for understanding” where the student has to solve a few problems correctly to complete the lesson.

Deliverables

- **One** working prototype of a single math lesson on fraction equivalence.
- The application must be a web-based app, runnable in a standard browser.
- A 1-2 minute demo video showcasing the conversational flow and the interactive fraction manipulative.

- A README file with instructions on how to run the application and a brief overview of your technical approach.

Requirements

- This application **must be built to run on an iPad** in a web browser. An iPad will be provided for testing.
- There are no restrictions on programming languages, frameworks, or tools. Use whatever you think is best to get a working demo in one week.